POWER BUDGET

Optical object-detection board • preliminary

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This is a preliminary sizing worksheet for the proposed optical object-detection board. The PIC/Curiosity Nano and miscellaneous loads are engineering allocations, not verified maximum currents. Consequently this page is not yet the completed maximum-current budget required for external design review. No measurements have been taken.

5 V load	Quantity	Current (mA)	Basis
PIC18F57Q43 Curiosity Nano	1	100.00	Planning allowance for board + MCU; verify operating mode and board circuitry
IR333-A emitter	1	20.00	Proposed ON-state current; do not use duty-cycle average for rail sizing
PN2222 base drive	1	2.00	Proposed GPIO current allowance; included in rail load
MCP6002 amplifier channels	2	0.34	2 × 170 µA quiescent maximum at datasheet test conditions; output loading covered below
Midpoint divider, 10k + 10k	1	0.25	5 V / 20 kΩ
Photodiode / feedback signal	1	0.02	20 µA assumed signal allocation; revise after optical tests
Indicator, pull-ups and spare analog loading	—	10.00	Provisional allowance; replace with actual supporting component values
Total planned 5 V load		132.61	Sum of entries above
25% sizing margin		33.15	0.25 × 132.61 mA
5 V rail design allocation		165.76	1.25 × 132.61 mA ≈ 0.166 A

Power sizing & remaining checks

Regulator and source sizing

With the allocation above, 5 V output power is 5 × 0.16576 ≈ 0.829 W. Allowing a provisional 10 mA for L7805CV ground current gives approximately 0.176 A at the 9 V input, or 1.58 W. The selected 9 V, 1 A adapter provides substantial current margin. Replace the 10 mA allowance with the applicable datasheet maximum across input, load and temperature before final review.

Thermal check

At nominal input, regulator dissipation is approximately (9 – 5) × 0.16576 + 9 × 0.010 ≈ 0.753 W. Using a 62.5 °C/W TO-220 junction-to-ambient figure gives an illustrative rise of 47 °C. At 25 °C ambient, this estimates about 72 °C junction temperature. This is a screening calculation, not thermal validation; adapter tolerance, enclosure, airflow and board mounting must be included, and a heatsink added if needed.

Regulator alternatives

The Component Selection page compares three linear regulators: L7805CV, LM317TG and LM2940CT-5.0/NOPB. L7805CV is preferred for the fixed 5 V / 9 V architecture and price. LM317 requires a divider and minimum-load check; LM2940 requires the specified output-capacitor ESR and offers little dropout advantage from 9 V. All three require a dissipation check.

Power-source alternatives and battery life

The source comparison includes Adafruit 9 V / 1 A, Pololu 9 V / 1 A and Pololu 9 V / 5 A adapters. A mains adapter is selected to satisfy the course architecture. Battery lifetime is not applicable to this configuration; no battery capacity or runtime is claimed.

Items needed for the completed budget

Obtain a documented worst-case MCU/board current for the selected clock, ADC, GPIO and debugger/power state; verify Curiosity Nano external-power wiring; replace indicator/pull-up allocations with schematic values; check maximum LED current with supply/resistor/transistor tolerances; include external bus loading; update regulator quiescent current and ambient assumptions; transfer the verified values into the course power-budget template.

MCP6002 datasheet · L7805CV datasheet · Selected 9 V adapter · Course power-budget requirements